

Axel Sanchez

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Education

University of Calgary, Bachelor of Science in Software Engineering – Calgary, Alberta September 2021 – April 2026

Relevant Coursework: Data Structures & Algorithms, Software Architecture, Operating Systems, Networked Systems, Database Management, Object-Oriented Software Design, Machine Learning Systems, Software Testing & Quality, Full Stack Web Development, Software Performance Evaluation, Discrete Mathematics

Experience

Software Engineer Intern, Microsoft – Vancouver, British Columbia May 2025 – August 2025

- Implemented custom **CUDA** All-Reduce kernels (one-shot, two-shot, user-buffers) for intra-node tensor parallelism on **NVIDIA H200** GPU clusters; integrated as a **TensorRT-LLM** plugin in **C++** with **NVLink/P2P** fallback
- Developed **BLAS** primitives (**GEMM**, **AXPY**, dot-product, reduction) in **C/CUDA** using tiled shared-memory, loop unrolling, and IPC buffers; benchmarked against **NCCL** and **cuBLAS** across **FP16/FP32**
- Engineered an **IPC-based user-buffers AllReduce** leveraging zero-copy peer access across intra-node **NVIDIA H200** GPUs, achieving up to **2× speedup** over **NCCL** on sub-2 MB payloads for **tensor-parallel LLM** inference; fused an **AllReduce + GEMM** **CUDA** kernel to overlap reduction with matmul

Cloud Developer Intern, Aptum – Calgary, Alberta May 2024 – April 2025

- Managed production **Kubernetes** infrastructure with **Terraform** and **Helm**; optimized **Jenkins CI/CD** pipelines to **double** deployment throughput
- Refactored legacy monolith into **gRPC-based microservices** using **Spring Boot** and **MySQL**, improving request latency by **66%** and enabling independent service scaling
- Authored DevOps runbooks, Helm chart documentation, and onboarding guides adopted across 3+ intern cohorts

Software Engineer Intern, Sigma Foodservice – Calgary, Alberta May 2023 – August 2023

- Integrated Copomex geolocation API with **C#/.NET** services across **10,000+** retail locations, improving address-resolution accuracy by **80%**
- Automated end-to-end regression testing with **Playwright** and **Jenkins CI/CD**, reducing manual QA effort by **98.75%** and accelerating release cycles
- Drove **Agile/Scrum** sprint reviews with product stakeholders to align engineering scope, acceptance criteria, and delivery timelines

Research Experience

Undergraduate Researcher, University of Calgary – Calgary, Alberta September 2025 – Present

- Designing **BridgeMart**, a cross-chain (Ethereum/Solana) data marketplace enabling trustless dataset listing, semantic discovery, and encrypted purchase via **Solidity** smart contracts, **LLM embeddings**, and **IPFS** off-chain storage
- Built privacy-preserving data-vending pipeline: client-side dataset encryption, on-chain integrity hashes, and smart-contract-gated decryption-key release (**hasAccess**) through a **FastAPI/Pydantic** backend
- Shipped a **RAG-style** dataset discovery service using **LLM embeddings** to encode tabular signatures and natural-language queries into a shared vector space for ranked similarity search; exposed a **REST API** consumed by **React/TypeScript + React Native** wallet-integrated frontends

Projects

ChiselWare IP Factory Platform (Capstone) – *React, TypeScript, FastAPI, MySQL, AKS* Rocksavage Technology Inc.

- Co-built a full-stack marketplace with a 6-engineer team enabling verified submission, certification, and monetized distribution of Chisel-based semiconductor IP cores; **React/Vite/TypeScript** frontend with **TanStack Query** and a **FastAPI/Python** backend using **SQLModel** over **MySQL**
- Engineered cloud deployment on **Azure Kubernetes Service** with **Terraform**-provisioned infrastructure, **Azure Container Registry**, and **Azure Key Vault** secret management; automated builds and deploys through **GitHub Actions** CI/CD
- Integrated **Stripe** checkout with webhook-driven access grants, **LinkedIn OIDC** authentication, automated **GitHub** repo provisioning, and a parameterized Chisel-to-Verilog compilation worker; validated with **pytest**, **Vitest**, and **Playwright** E2E suites

Leadership Experience

Founding Engineer, Agrivanna – Calgary, Alberta September 2024 – April 2025

- Architected **RESTful** microservices in **Go** and **Spring Boot** for crop-health APIs; containerized with **Docker** and orchestrated on **Kubernetes**
- Productionized a **PyTorch** time-series yield-forecasting model consuming **Kafka** streams; served predictions via **FastAPI** endpoints with CI/CD via **GitHub Actions**

VP Software Development, Student Energy UCalgary – Calgary, Alberta September 2023 – April 2024

- Led a 6-engineer team to launch a **Next.js/TypeScript** platform, increasing chapter engagement by **70%**; directed **AWS**, **Terraform**, and **GitHub Actions** CI/CD infrastructure
- Owned hiring pipeline: defined evaluation rubrics, conducted 15+ interviews, and authored executive succession documentation to ensure continuity

Skills

Languages: Python, Java, C, C++, C#, Go, Rust, JavaScript, TypeScript, SQL, Solidity, Dart

Frameworks & Libraries: Spring Boot, .NET, Django, FastAPI, Node.js (Express), React, Next.js, React Native, Flutter, GraphQL

AI & ML: Generative AI, LLMs, AI Agents, RAG, Prompt Engineering, PyTorch, Transformers, OpenAI API, LangChain, TensorRT-LLM, CUDA, Scikit-learn, MCP

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Terraform, Helm, Jenkins, GitHub Actions, CI/CD, Nginx

Databases & Messaging: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Kafka, RabbitMQ, IPFS

Concepts: Distributed Systems, Microservices, REST APIs, gRPC, System Design, Data Structures & Algorithms, OOP, Agile/Scrum, Unit Testing, MLOps, IaC

Certifications: Microsoft Certified Azure Fundamentals (2025), AWS Certified Cloud Practitioner (2024)